GRADE 9

Computational Thinking

National Level Examination

SUBJECT CODE			

Total Questions: 50 **Total Marks: 50** **Time: 75 Minutes**

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED TO DO SO

- **Carefully Check Your Name, Class, Roll Number, And Other Required Details On The OMR Answer Sheet.**
- **Do Not Forget To Sign The OMR Sheet. Also Write Your Roll No. On The Question Paper Booklet.**
- **Use Only Black/Blue Ball Pen And Hb Pencil To Fill The OMR Answer Sheet.**
- **All Questions Are Compulsory And Carry One Mark Each.**
- **Section A: Abstract & Spatial Thinking (10 Questions); Section B: Pattern Recognition (10 Questions); Section C: Decomposition, Data & Constraints (10 Questions); Section D: Algorithmic Thinking & Logical Flow (10 Questions); Section E: Ai Literacy & Olympiad Challenge (10 Questions).**
- **Only OMR Answer Sheets Will Be Evaluated.**
- **No Negative Marking For Incorrect Answers.**
- **You May Use The Space Provided On The Last Page For Rough Work (If Needed).**
- **Digital Devices Like Calculators, Smart Watches, Or Mobile Phones Are Not Allowed.**
- **After The Exam, Submit Both The OMR Sheet And The Question Paper To The Invigilator.**

SAMPLE

SECTION A – ABSTRACT & SPATIAL THINKING

Q1. What is the decimal value of the binary number 110101?
A. 49
B. 51
C. 53
D. 55

Q2. The point (−4, 3) is rotated 90° clockwise about the origin. What are its new coordinates?
A. (−3, −4)
B. (3, 4)
C. (−3, 4)
D. (4, 3)

Q3. A 4 × 4 × 4 cube is painted on all outer faces and cut into unit cubes. How many unit cubes have exactly two painted faces?
A. 16
B. 20
C. 24
D. 32

Q4. All objects in set A are in set B, and no object in set B is in set C. Which statement must be true?
A. Some A are C
B. No A is C
C. All C are A
D. A and C are identical

Q5. Which binary number represents the decimal number 73?
A. 1000001
B. 1001001
C. 1010001
D. 1011001

Q6. In a group of 80 students, 45 study Python, 30 study AI and 15 study both. How many study neither?
A. 10
B. 15
C. 20
D. 25

Q7. The binary no.11010 equals to
A. 56
B. 57
C. 58
D. 60

Q8. What is the binary value of decimal 46?
A. 101010
B. 101110
C. 110010
D. 111000

Q9. A 5 × 5 × 5 cube is painted on all six faces and cut into unit cubes. How many unit cubes have no painted face?
A. 8
B. 27
C. 36
D. 64

Q10. The point (−2, −5) is reflected across the x-axis. What are its new coordinates?
A. (2, −5)
B. (−2, 5)
C. (5, −2)
D. (−5, 2)

SECTION B – PATTERN RECOGNITION

Q11. Find the next number: 2, 5, 12, 27, 58, ___.
A. 116
B. 119
C. 121
D. 123

Q12. Find the next number: 3, 8, 15, 24, 35, ___.
A. 44
B. 46
C. 48
D. 50

Q13. Find the next number: 1, 4, 13, 40, 121, ___.
A. 242
B. 343
C. 364
D. 366

Q14. Find the next number: 6, 12, 11, 22, 21, 42, 41, ___.
A. 80
B. 81
C. 82
D. 84

Q15. A rule maps n to n(n + 1). If 2 → 6, 3 → 12 and 5 → 30, what is 8 → ?
A. 64
B. 70
C. 72
D. 80

Q16. B, C, E, G, K, M, Q correspond to prime-numbered alphabet positions. Which letter comes next?
A. R
B. S
C. T
D. U

Q17. Find the next number: 5, 11, 23, 47, 95, ___.
A. 181
B. 189
C. 191
D. 193

SAMPLE

SECTION B – PATTERN RECOGNITION (CONTINUED)

Q18. Find the next number: 2, 3, 6, 11, 18, 27, ___.
A. 36
B. 38
C. 40
D. 42

Q19. Find the next hexadecimal value: A, C, F, 13, 18, ___. (values shown in base 16)
A. 1C
B. 1D
C. 1E
D. 1F

Q20. Find the next number: 1, 1, 2, 3, 5, 8, 13, ___.
A. 18
B. 20
C. 21
D. 22

SECTION C – DECOMPOSITION, NUMBERS & DATA

Q21. A quantity increases by 20% and then decreases by 20%. If it starts at 500, what is its final value?
A. 480
B. 490
C. 500
D. 520

Q22. The mean of 14, 18, 21, 27 and x is 22. What is x?
A. 28
B. 30
C. 32
D. 34

Q23. A dataset of 1200 records is divided 70% training, 15% validation and 15% testing. How many records are in the test set?
A. 150
B. 180
C. 210
D. 240

Q24. A 64 × 64 grayscale image stores 8 bits per pixel. Ignoring metadata, how many bytes are needed?
A. 2048
B. 4096
C. 8192
D. 32768

Q25. Red and blue tokens are in the ratio 3 : 5, with 160 tokens total. If 25% of the blue tokens are changed to red, what is the new ratio red : blue?
A. 15 : 17
B. 17 : 15
C. 7 : 5
D. 5 : 3

Q26. A model classifies 500 cases: 360 of 400 Class-A cases and 70 of 100 Class-B cases are correct. What is its overall accuracy?
A. 82%
B. 84%
C. 86%
D. 88%

Q27. A fair die is rolled once. What is the probability of getting a prime number?
A. 1/3
B. 1/2
C. 2/3
D. 5/6

Q28. A dataset has values 4, 5, 5, 6, 10. What is the median?
A. 5
B. 6
C. 6.5
D. 10

Q29. A 256 × 256 RGB image uses 24 bits per pixel. Ignoring metadata, how many bytes are needed?
A. 65,536
B. 131,072
C. 196,608
D. 262,144

Q30. If 18 of 60 predictions are wrong, what is the model's accuracy?
A. 60%
B. 65%
C. 70%
D. 75%

SECTION D – ALGORITHMIC THINKING & LOGICAL SEQUENCING

Q31. Start with x = 4. For i = 1, 2, 3, update x = x × i + 1. What is the final value of x?
A. 30
B. 32
C. 34
D. 36

Q32. Apply this rule three times: if x is even, x = x/2; otherwise x = 3x + 1. Start with x = 5. What is x after three steps?
A. 2
B. 4
C. 8
D. 16

Q33. Variables have a = 7 and b = 11. Execute temp=a; a=b; b=temp. What are the final values?
A. a=7, b=11
B. a=11, b=7
C. a=11, b=11
D. a=7, b=7

Q34. A nested loop runs j from 1 to i for each i from 1 to 4. How many times does the inner action run in total?
A. 8
B. 9
C. 10
D. 16

SAMPLE

SECTION D – ALGORITHMIC THINKING & LOGICAL SEQUENCING (CONTINUED)

Q35. A weighted network has edges A-B=2, A-C=6, B-C=1, B-D=5 and C-D=2. What is the minimum cost from A to D?
A. 4
B. 5
C. 6
D. 7

Q36. An average algorithm sets sum=0 inside the loop before adding each new value. What is the correct fix?
A. Set sum=0 once before the loop
B. Set sum=1 inside the loop
C. Divide before adding
D. Remove the loop

Q37. Set s = 0. For i = 1 to 6, if i is odd add i^2 to s; otherwise add i. What is s?
A. 38
B. 41
C. 44
D. 47

Q38. A binary search is used on a sorted list of 64 items. What is the maximum number of halving steps needed to reduce to one item?
A. 5
B. 6
C. 7
D. 8

Q39. A shortest grid path needs 4 Right moves and 3 Up moves. How many shortest paths are possible?
A. 20
B. 30
C. 35
D. 42

Q40. A loop should run for i = 1 to 10, but the condition is written i < 10. How many times does it run?
A. 8
B. 9
C. 10
D. 11

SECTION E – OLYMPIAD CHALLENGE

Q41. An AI predicts a student's numerical score from previous performance data. Which technique is most suitable?
A. Classification
B. Regression
C. Clustering
D. Encryption

Q42. A classifier marks 50 items as positive; 40 of those are actually positive. What is its precision?
A. 70%
B. 75%
C. 80%
D. 90%

Q43. An AI groups documents by similarity without predefined labels. Which method is being used?
A. Classification
B. Regression
C. Clustering
D. Sorting only

Q44. A medical AI is trained on data that under-represents one population. What should be checked most carefully before deployment?
A. Font size
B. Bias and fairness across groups
C. Screen brightness
D. File extension

Q45. Which practice best protects privacy when collecting data for an AI system?
A. Collect every possible detail
B. Collect only necessary data with informed consent
C. Share raw data publicly
D. Keep passwords in the dataset

Q46. A 4-digit code uses 1, 2, 3 and 4 once each. The first digit is twice the second, first + last = 5, third > last, and third < first. What is the code?
A. 2143
B. 2413
C. 4231
D. 4312

Q47. A model performs very well on training data but poorly on new data. This is most likely:
A. Clustering
B. Overfitting
C. Encryption
D. Compression

Q48. Which metric measures the fraction of actual positives that a classifier correctly finds?
A. Precision
B. Recall
C. Mean
D. Variance

Q49. Which AI domain is central to understanding and generating human language?
A. Computer Vision
B. Natural Language Processing
C. Robotics only
D. File Management

Q50. A generative-AI answer sounds confident but contains false facts. What is the best response?
A. Accept it because it sounds certain
B. Verify important claims using reliable sources
C. Share it immediately
D. Remove all source citations

SAMPLE